

Object Oriented Programming (CSC166)

Lab Sheet 4

1. Write a program that defines a class "Distance" with private data members feet and inches. Write a conversion constructor that converts a basic float value (representing total distance in feet, e.g., 5.75) into a Distance object, splitting it into whole feet and inches.

*Hint: [Distance(float f) → conversion constructor that takes a float value f (total feet), sets feet = (int)f, and sets inches = (f - feet) * 12*

displayDistance() → prints the distance in the form feet'-inches"]

2. Write a program that defines a class "Distance" with private data members feet and inches. Overload the type conversion operator to convert a Distance object into a basic float value representing the total distance in feet (i.e., feet + inches/12).

Hint: [Distance(int f, int i) → parameterized constructor

operator float() → conversion operator (no return type specified, no arguments) that returns feet + (inches / 12.0) as a float

Test by assigning a Distance object directly to a float variable, e.g., float d = distObj;]

3. Write a program that defines two classes "Kilometers" and "Miles", where Kilometers has a private data member km, and Miles has a private data member miles. Write a conversion constructor inside the Miles class that converts a Kilometers object into a Miles object (assume 1 km = 0.621371 miles).

Hint: [Kilometers(float k) → constructor to initialize a Kilometers object

*Miles(Kilometers k) → conversion constructor inside the Miles class that computes miles = k.km * 0.621371*

Use public getter method in Kilometers to access km

displayMiles() → prints the distance in the form "Miles: miles"

displayKilometers() → prints the distance in the form "Kilometers: km"]

4. Write a program that defines two classes "Dollars" and "Rupees", where Dollars has a private data member amount, and Rupees has a private data member amount. Overload the type conversion operator inside the Dollars class to convert a Dollars object into a Rupees object (assume 1 Dollar = 135 Rupees).

Hint: [Dollars(float amt) → parameterized constructor

*operator Rupees() → conversion operator inside Dollars that returns a Rupees object with amount = amount * 135*

displayRupees() → prints the amount in the form "Rs. amount"

Rupees must be declared/defined before the Dollars class, or forward-declared, since Dollars needs to know about Rupees]